

Yiling Fan

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EDUCATION

University of Cambridge, Murray Edwards College

Cambridge, UK

Degree: BA (Hons); MEng in Engineering

10/2023 – 06/2027(Expected)

Academic Result: BA overall Class I (First Class)

Selected Coursework: Systems and Control, Signals and Systems, Statistical Signal Processing, Integrated Digital Electronics, Information Theory and Coding.

INTERNSHIP EXPERIENCE

Geely Auto Research Institute

Ningbo, China

Summer Intern, Autonomous Parking Department

07/2025 – 09/2025

- Utilized TSMaster to control vehicle chassis and measured turning radii at various steering wheel angles across different vehicle models.
- Studied MATLAB to develop PID algorithms and built state-space and Simulink simulation models for a two-degree-of-freedom spring oscillator.
- Studied path prediction algorithms (Adaptive Lookahead Pure Pursuit Algorithm) for precise lateral control in autonomous parking.
- Analyzed autonomous parking test issues using MATLAB and PlotJuggler, evaluating vehicle speed, steering angle, heading angle deviation, and lateral deviation to identify abnormal steering or braking behaviors, support root-cause investigation, and inform subsequent code improvements.

Summer Intern, New Energy Section Energy Control Department

07/2024 – 08/2024

- Developed weekly BI dashboards to visualize post-sale vehicle issue data, including error-cause distributions, issue resolution rates, and other operational metrics, helping team members and managers monitor weekly progress, identify recurring issues, and prioritize unresolved cases.
- Used Pyecharts library in Python to plot battery error voltage vs. time graphs, merging data from multiple vehicles.
- Assisted in an AI Agent project including learning AI orchestration logic, prompt engineering, and how to categorize user inputs to search specific knowledge bases and generate solutions.
- Awarded “Outstanding Individual” and “Outstanding Team” during onboarding.

ENGINEERING & ACADEMIC PROJECTS

Visual-Inertial Odometry (VIO) on Embedded Hardware

Cambridge, UK

Advised by Professor Claudel

03/2026 – Present

- Developed and evaluated autonomous localization and multi-sensor fusion schemes to estimate real-time 2D position and heading angle for a Picar-x wheeled robot.
- Calibrated the non-linear kinematics of the robot using linear regression and the least squares, fitting the linear velocity parameters and angular velocity parameter.
- Implemented a complementary filter with a weight coefficient to fuse the kinematic model with IMU gyroscope measurements, resolving the severe drift issue associated with accelerometer double integration.
- Configured a Raspberry Pi embedded environment and resolved camera FPC cable hardware connection issues; utilized OpenCV to capture remote image streams, extract ORB features, and perform two-frame matching.
- Executed ORB-SLAM2 to estimate camera pose matrices, and initiated 2D LiDAR mapping and localization using slam_toolbox to design a sensor fusion scheme mitigating monocular scale ambiguity.

Optical Fiber Link Project

Cambridge, UK

Advised by Professor Tim Wilkinson

05/2026 – 06/2026

- Built a 2 Mbit/s plastic optical fiber digital communication link over 20 meters under intense electromagnetic interference, successfully restoring signals to standard 5V CMOS/TTL levels.
- Developed a feedforward Automatic Gain Control architecture combining a peak envelope detector and an OP37 integrator to drive a BS170 NMOSFET, achieving dynamic attenuation and preventing amplifier saturation under short-distance high-power inputs.
- Designed a precision hysteresis comparator using a MAX941, establishing a 426 mV hysteresis window to eliminate noise-induced false triggering, and cascaded a 74LS14 Schmitt trigger to compress edge transition times to under 200 ns.
- Created an audio output isolation interface with a 10:1 voltage divider and an LT1252 unity-gain follower, successfully decoding and playing real CD digital audio streams through an 8Ω speaker.

Industrial Evaporator Multivariable Control System

Cambridge, UK

Advised by Professor Fulvio Forni

05/2026 – 06/2026

- Designed a multivariable control system for an industrial evaporator to regulate product concentration under safety constraints of separator level and operating pressure, building a non-linear Simulink model and extracting

state-space matrices via MATLAB linmod.

- Developed decentralized PI controllers using frequency response analysis and Ziegler-Nichols tuning, implementing a Clamping Anti-windup algorithm to eliminate steady-state deviation and overshoot under actuator saturation.
- Designed a one-dimensional gain scheduling controller based on product concentration to manage non-linear dynamics, reducing settling time by 2.6 times from 400 down to 150 minutes and cutting coupling disturbances by 50% during operating point transitions.
- Formulated an augmented state-space state feedback controller and a Linear Quadratic Regulator using Bryson's rule to achieve optimal multivariable decoupling and eliminate steady-state tracking errors.

Integrated Design Project (IDP) for a Robot

Cambridge, UK

Group Project

11/2024 – 12/2024

- Collaborated in a team of 6 to build an autonomous line-following robot for sorting magnetic boxes, leading the electrical sub-team to design the adapter board circuit and integrate it with the PicoBot board.
- Calibrated multiple sensors including ultrasonic and line detectors, performing precise soldering and short-circuit testing to ensure seamless communication across mechanical, electrical, and software systems.

Flight Computer PCB Design, Cambridge Spaceflight Society

Cambridge, UK

Group Project

10/2024 – 12/2024

- Contributed to a flight computer design project focused on developing a real-time pressure sensor.
- Designed the Printed Circuit Board (PCB) using KiCad, managing component placement and routing.

Structural Design Project (Aluminum Bridge)

Cambridge, UK

Advised by Professor Julian Allwood

02/2024 – 03/2024

- Designed and manufactured a Warren truss aluminum bridge to withstand a 3.5kN load without deformation and fail at 7.0kN, utilizing 3D force calculations to optimize bar and bolt sizing and producing detailed manual engineering drawings.

Engineering Coursework & Lab Projects, Cambridge, UK

- **Mechanical Design (SolidWorks):** Designed and simulated a gravity-powered rollercoaster; developed 3D models, assemblies, and animations, and performed motion analysis to meet design criteria.
- **Electrical Circuits:** Built high/low-pass filters and resonant circuits; utilized LTSpice and PicoScope to measure time constants and resonant frequencies.
- **Mars Lander (C++):** Modeled dynamics, simulated orbital mechanics, and developed an autopilot system based on control theory.
- **Microcontroller Programming:** Programmed a temperature control system in C++ and machine code, managing memory, interrupts, and peripheral communication.
- **Flood Warning System (Python):** Coded a real-time warning system collecting and processing data from monitoring stations.

HONORS

- Mary Dyson Scholarship, 11/2024 & 12/2025
 - Awarded by the James Dyson Foundation to top engineering students.
- Global Gold & Best in School (Full mark, Top 10%), Senior Mathematical Challenge, 2022.
- National High Distinction (Top 5%), Australian Mathematics Competition, 2022.
- Selected Participant, National Talent Program in Physics, 12/2020 – 12/2021
 - Chosen by the Chinese Ministry of Education for an early STEM enrichment program at Zhejiang University, gaining introductory exposure to materials synthesis and superconductivity.

EXTRACURRICULAR ACTIVITIES

Cambridge University Women's Basketball Club, *Team Player & Safety and Welfare Officer*

10/2023 – Present

High School Women's Basketball Club, *Vice President & Co-Founder*

09/2020 – 06/2023

SKILLS

Programming & Software: Python, C++, MATLAB, MySQL

Simulation & Control: Simulink, PlotJuggler, TSMaster

Design & EDA: SolidWorks, KiCad, LTSpice

Languages: Chinese (Native), English (Proficient, IELTS 8.0)

Others: Piano National Level 10